

Benjamin Pachter

Principal AI & Data Architect · Full-Stack LLM Engineer

01 SUMMARY

Principal-level AI & data architect who ships production ML/LLM platforms end to end — data architecture and pipelines, model evaluation, and scalable serving (RAG / vector search, multi-model routing, GPU inference). At Duke Energy, architected a geospatial capacity-modeling platform screening 100+ GW of transmission load, supported 150+ hyperscaler data-center site analyses, and cut load-study turnaround from 2–4 weeks to same-day (~200 engineering hours saved / month). Own platform roadmap and technical standards across five internal organizations, translating ambiguous business needs into production-grade data / AI systems. Full-stack application developer by hobby.

AI / ML Architecture

LLMOps / MLOps

Data Architecture

ETL / ELT

APIs & Microservices

AWS

Docker / Kubernetes

CI / CD

Agentic AI & RAG

Evals & Observability

02 TECHNICAL SKILLS

Agents & LLMOps	LangGraph, LangChain, CrewAI, MCP (Model Context Protocol), multi-agent orchestration, tool-calling loops, structured outputs, context engineering, DSPy
RAG & Knowledge Graphs	Grounded RAG with citations, agentic RAG, hybrid search, reranking, GraphRAG; knowledge-graph construction, ontology design, LLM entity / relationship extraction; embeddings, ChromaDB, pgvector, Qdrant
Evals & Observability	Evals & LLM-as-judge (local + cloud), RAGAS, LangSmith, tracing (Langfuse, Phoenix), app monitoring (Sentry), guardrails, prompt-injection defense (OWASP LLM Top 10)
Serving & Tuning	vLLM, SGLang, Ollama (local RTX 4090), quantization (GGUF / AWQ), prompt caching, multi-model routing & fallback; fine-tuning — LoRA / QLoRA, DPO, SFT, RLHF, Hugging Face
Foundation Models	Claude (Anthropic SDK), GPT-5.x, Gemini, Gemma, Llama, Qwen, DeepSeek, Mistral
ML / Deep Learning	PyTorch, TensorFlow, Transformers, XGBoost / LightGBM, CNNs, time-series forecasting, feature engineering; experiment tracking (MLflow, W&B)
Backend & APIs	Python (FastAPI), Node.js, Express, TypeScript, REST & SSE streaming, WebSockets, microservices; JWT / RBAC auth, 2FA (TOTP), rate limiting, multi-tenant SaaS
Data Engineering	SQL, Pandas, NumPy, PySpark, Airflow, ETL / ELT, multi-source connector ingestion, scheduled pipelines (cron), canonical schema design; Java, C++, C#, Scala, Bash
Cloud & Infra	AWS (S3, Lambda, Glue, Athena, Redshift, SageMaker, Bedrock), Docker, Kubernetes, CI/CD, Railway / Nixpacks, Cloudflare R2, PostgreSQL, SQLite, CUDA
Geospatial	ArcGIS Pro, Python toolboxes, spatial raster modeling, IDW interpolation, transmission capacity analysis, PSS/E integration
Frontend & Dataviz	React, TypeScript, Vite, MUI, Emotion, TanStack Query; Three.js / react-three-fiber, D3 (force-directed graphs), Plotly, MapLibre GL, lightweight-charts, Recharts

Duke Energy

Sep 2024 – Present

Senior Business Analyst

Charlotte, NC · Remote

- Architected and delivered a spatial transmission-capacity modeling platform (ArcGIS Pro + Python toolbox) generating a continuous MW-headroom raster (IDW interpolation on tower-level TrLim); screens 100+ GW of large-load requests and has supported 150+ hyperscaler data-center site analyses across a 6-state territory.
- Designed CIM-aligned, AI-ready data architecture and integration patterns making grid and asset data interoperable for downstream analytics, ML, and future LLM applications enterprise-wide.
- Built prompt-driven and agentic workflows (LangChain) automating site screening, feasibility narratives, and decision support — integrating structured enterprise data with LLM reasoning to cut manual analysis loops.
- Reduced PSS/E load-hosting study turnaround from 2–4 weeks to same-day by automating pre-screening and eliminating infeasible sites before engineering analysis; saved ~200 engineering hours / month.
- Engineered reusable production data pipelines integrating AWS, SAP S/4HANA, Salesforce (Apex / Flows), and ArcGIS across 5 internal organizations; automated operational reporting and a major industrial customer's monthly billing pipeline.
- Own platform roadmap, technical standards, and stakeholder alignment across transmission planning, economic development, and operations.

New York Mortgage Trust

Jan 2022 – May 2024

Acquisitions Analyst

Charlotte, NC

- Built multi-source ingestion pipelines (Python, SQL) integrating Yardi and public-data APIs (FRED, FDIC, BLS, Census); delivered forecasting models (TensorFlow, XGBoost) for rent, occupancy, and cap-rate prediction to support institutional underwriting.
- Deployed CNN-based document parsing and BERT / Transformer NLP pipelines for automated document understanding and sentiment analysis; migrated legacy workflows to cloud-native AWS / Azure services.
- Delivered ML-driven risk scoring and executive analytics across a \$7B+ residential mortgage portfolio; implemented Salesforce automations (Apex triggers, Flow) to streamline operations.

Deloitte

Jan 2020 – Jan 2022

Risk & Financial Advisory Analyst

Seattle, WA

- Designed automated ETL pipelines (Python, AWS) and ML models (Random Forest, GBM, ARIMA, GARCH, LSTM) for portfolio risk forecasting and bond-volatility prediction across finance, energy, and defense clients.
- Built NLP pipelines for financial sentiment analysis across news and social media; supported SAP S/4HANA + Azure SQL migrations and delivered audit-automation frameworks for Fortune 500 clients.

BMW Manufacturing

Jan 2018 – Dec 2018

Supply Chain Planning Co-op

Greer, SC

- Drove ~\$13M annual transportation-cost reduction by optimizing SAP reorder algorithms; built time-series forecasting models and predictive dashboards (Python, Power BI) for supply-chain disruption risk across G05–G09 vehicle platforms.

04 SELECTED PROJECTS

thessa.space

2025 — Present

Founder & Lead Engineer

Financial-intelligence terminal for the space & physical economy — defense, semiconductors, energy, and critical minerals — turning SEC filings, federal contracts, launches, options flow, and congressional trades into one ranked feed over a queryable knowledge graph.

- **Hybrid model fleet** — a role-specialized swarm of local open-source agents (Qwen 2.5 / 3, Gemma 2, Llama 3.x, nomic-embed, faster-Whisper) on a 24 GB RTX 4090 runs research, extraction, embedding, transcription, and first-pass adjudication at ~\$0; cloud Claude (Haiku / Sonnet) drives the user-facing analyst and escalated verification — **local-first, cloud-as-fallback**, budget-capped.
- **Autonomous knowledge graph** — ~11,000 companies, ~27K supply-chain edges, and 78K+ federal contracts from 15+ data connectors; an agentic flywheel proposes edges and an independent cited-web-search adjudicator verifies them, every edge shipping with sources and a confidence score.
- **Platform** — real-time React / TypeScript terminal (WebGL globe, D3 graphs, Plotly) on a Node / Express API; grounded Claude analyst with row-level citations and tool-loops; commercial metered REST API with org / seat tenancy and JWT + 2FA; shipped on Docker / Railway with Sentry observability.

Stack — React · TypeScript · Vite · MUI · react-three-fiber · D3 · Plotly · MapLibre · Node / Express · Anthropic SDK · Ollama · SQLite · Postgres · Docker · Railway · Cloudflare R2 · Sentry · SSE · 15+ data connectors

Entropy

bpachter.github.io/entropy

An interactive, scroll-driven history of thermodynamics — seven distinctly-themed chapters, each pinning a live, physically-accurate simulation you can manipulate beside the narrative: a working Carnot cycle, Maxwell's demon, a Shannon-entropy meter, and Hawking black-hole evaporation.

- **Hand-written physics engine** — a framework-agnostic, zero-dependency hard-sphere gas (uniform-grid collision broad-phase, coarse-grained entropy metrics) rendered with instanced WebGL particles; every interactive computes real quantities ($S = k \log W$, Stokes–Einstein, Landauer's $kT \ln 2$, Bekenstein–Hawking), with per-visual IntersectionObserver gating for smooth mobile performance.

Stack — React 18 · TypeScript · Vite 6 · react-three-fiber / three.js · framer-motion · Canvas 2D + WebGL

05 EDUCATION

University of North Carolina at Charlotte

Charlotte, NC

BSBA — Operations & Supply Chain Management and Management Information Systems

06 CERTIFICATIONS

IBM AI Engineering Professional Certificate

LLMs, Transformers, LoRA / QLoRA, RLHF, RAG, fine-tuning; PyTorch, TensorFlow, Hugging Face · 13 courses

Microsoft Power BI PL-300 Data Analyst Associate

ETL, DAX, data modeling, Microsoft Fabric deployment

SAS Statistical Business Analyst

Regression modeling, statistical analysis, hypothesis testing

DeepLearning.AI / Stanford — ML Specialization

Supervised / unsupervised learning, advanced algorithms, reinforcement learning

Architecting Solutions on AWS

Lambda, Glue, S3, Athena, Redshift, SageMaker